

THIN PLATE SMOOTHING SPLINE INTERPOLATION OF DAILY RAINFALL FOR NEW ZEALAND USING A CLIMATOLOGICAL RAINFALL SURFACE

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ABSTRACT

This study presents a method for estimating daily rainfall on a 0.05° latitude/longitude grid covering all of New Zealand for the period 1960–2004 using a second order derivative trivariate thin plate smoothing spline spatial interpolation model. Use of a hand-drawn (and subsequently digitised) mean annual rainfall surface as an independent variable in the interpolation is shown to reduce the interpolation error compared with using an elevation surface. This result is confirmed when long-term average annual rainfall data, derived from the daily interpolations, are validated using long-term river flow data. Copyright © 2006 Royal Meteorological Society.

KEY WORDS: daily rainfall; spline interpolation; New Zealand

1. INTRODUCTION

As New Zealand is located between the latitudes of 34 and 47°S , it lies between two important features of the global circulation: the band of subtropical anticyclones to the north, and the disturbed westerly belt to the south. New Zealand's rainfall is strongly influenced by these features. When high-pressure systems dominate the circulation, rainfall over New Zealand tends to be relatively low. Conversely, when the country is embedded in the westerlies rainfall is relatively high, especially in the exposed western regions, due to the frequent passage of frontal systems and depressions (Tait and Fitzharris, 1998).

The orography of New Zealand is shown in Figure 1. The country is long and narrow, with high and rugged relief on both islands spanning a distance of some 1300 km. The Southern Alps, which trend southwest to northeast along the length of the South Island, provide a significant barrier to the mean west to southwest airflow (Tomlinson, 1976), as do most of the country's other mountain ranges and volcanic peaks. The average precipitation pattern for New Zealand shows a strong west-to-east gradient, particularly in the South Island, which reflects the impact of the country's orography on rainfall. Annual rainfall totals are typically about 7500 mm on the west coast of the South Island, whereas east of the Southern Alps, in the rain shadow, annual rainfall totals are on average about 500 mm, increasing to about 800 mm on the east coast.

On a daily time scale, southerly outbreaks bring significant precipitation (often falling as snow at elevations above 500 m in winter) to southern and eastern areas of both islands, while summer heating of inland and eastern regions sometimes generates sufficient convection to trigger cumulus development and showers. Northwesterly and northerly airstreams, carrying moisture-laden air from the subtropics onto New Zealand, commonly produce heavy rainfalls in the north and west of both islands and dry föhn conditions in eastern areas. Northern New Zealand is also subject to heavy rainfall events associated with the passage of ex-tropical

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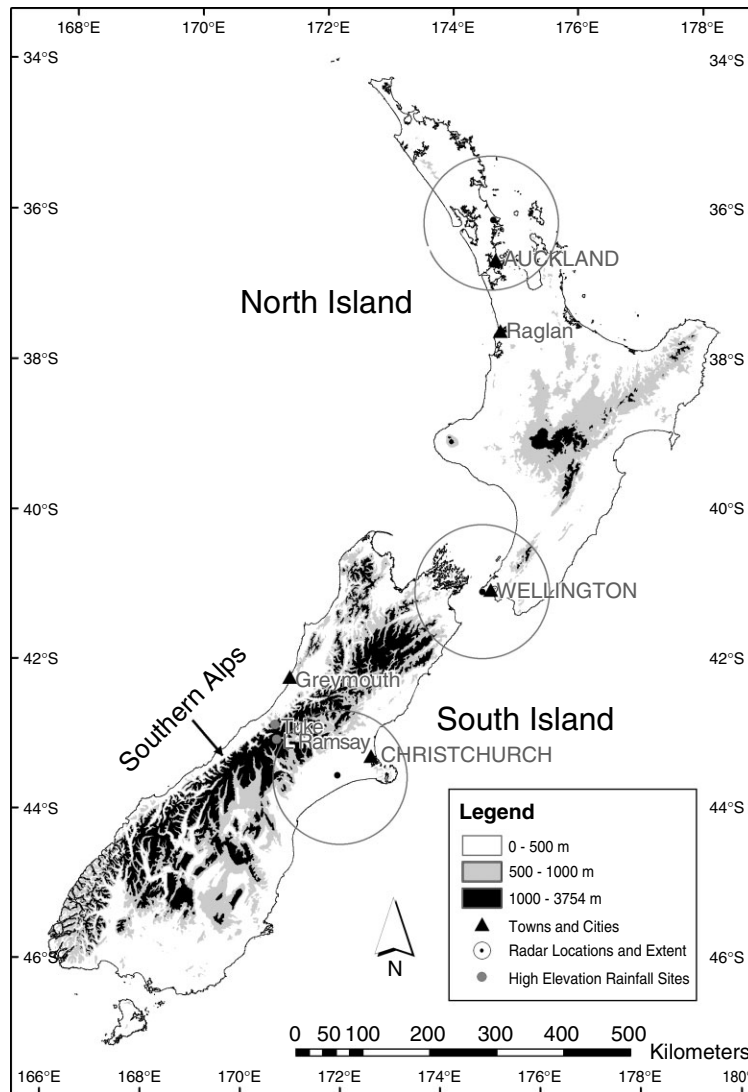


Figure 1. Map of New Zealand showing land above 500 m and 1000 m elevation, the three radar locations and their coverage, and locations of towns and cities mentioned in the text

depressions. Overall, the combination of location, orography, and season results in a distinctively regionalised rainfall regime (Salinger, 1980).

Spatial estimation of daily rainfall is an important issue in New Zealand, particularly for hydrological models and their applications. In recent years, hydrological models such as SPARROW (Schwarz *et al.*, 2001), have been used to provide information about the effects of land-use change on water quality in rivers. In New Zealand such information has increasingly been used to decide the fate of resource consent applications where degradation of waterways is an issue. Given the sensitive and important nature of such decisions, it is critical that the hydrological models and their key inputs be as accurate as possible.

Although there is a relatively dense network of climate stations in New Zealand (currently there are about 680 open stations measuring rainfall), the majority of these stations, 93%, are at elevations below 500 m above mean sea level (amsl). Thus, there are many mountainous areas and many river catchments where daily rainfall observations are not recorded. Hydrological models for these catchments are forced to rely on rainfall data from outside of the catchments, which introduces additional uncertainty into the calculations.

This study presents a method for estimating daily rainfall on a 0.05° latitude/longitude grid (approximately 4.8 km grid resolution (average of 4.2 km in latitude, 5.5 km in longitude, giving 23 km^2 grid cells)) covering all of New Zealand for the period 1960–2004 for the purpose of input into and testing of hydrological models. The approach uses a spatial interpolation methodology, incorporating two location variables and a mean annual rainfall variable. The paper is structured as follows. Section 2 compares four different methods of spatial estimation of rainfall (numerical weather prediction (NWP) models, radar-based methods, satellite-based methods, and interpolation of surface observations) and assesses their applicability for estimating long time series of historic daily rainfall in New Zealand. Section 3 presents some results from an initial exploratory data analysis. Section 4 describes and evaluates daily rainfall interpolations from two different parameterisations (bivariate and trivariate) of a thin plate smoothing spline model. Section 5 introduces and evaluates spline interpolations of daily rainfall using a mean annual rainfall surface, rather than elevation, as an independent variable in the model. Discussion and conclusions follow in Section 6.

2. SPATIAL ESTIMATION OF DAILY RAINFALL

There are four main methods of estimating daily rainfall at locations for which there are no surface observations. These are NWP models, radar data-based methods, satellite data-based methods, and surface data spatial interpolation methods. In this section, these four approaches are discussed and assessed for the purpose of generating a long time series (1960–2004) of historic daily rainfall estimates for all of New Zealand on a 0.05° latitude/longitude grid.

2.1. Numerical weather prediction models

With the existence of complete global reanalysis data sets that can provide lateral boundary conditions for numerical weather models, it is now technically feasible, albeit computationally expensive, to generate several years of daily rainfall estimates at high spatial resolution (e.g. horizontal grid spacing (or Δx) = 5 km, similar to 0.05° at mid-latitudes) for all of New Zealand. However, with a grid spacing of 5 km, numerical models can really effectively resolve only features that have a scale of 20–30 km. This is because of schemes that filter out grid-scale (4–6 km) and smaller features in the orography (Webster *et al.*, 2003). Verification studies such as by Ebert *et al.* (2003) have shown that while numerical quantitative precipitation forecast (QPF) fields give reasonable volume estimates, they are much smoother than observed gridded analyses.

While the study by Ebert *et al.* (2003) was for relatively coarse resolution models ($\Delta x > 40 \text{ km}$), the same tendency has been noted in New Zealand from a comparison of histograms at several locations for both three-hourly and daily rainfall estimates from mesoscale model (regional atmospheric modelling system (RAMS) $\Delta x = 20 \text{ km}$ and $\Delta x = 5 \text{ km}$) simulations. Here, a tendency for too many light rainfall events ($< 1 \text{ mm/day}$) and too few heavier rainfall events ($> 5 \text{ mm/day}$) was determined (Figure 2). In order to effectively resolve features on a 0.05° latitude/longitude grid and to avoid having fields too smooth, a grid spacing of about 1 km would be necessary. This would be 125 times more computationally expensive than a sequence of 5 km simulations (which are already expensive enough).

Hence, despite the potential for using numerical weather models to estimate long time series of daily rainfall at a high spatial resolution over New Zealand, the practical constraints caused by the large computational requirement make this approach inappropriate. Other disadvantages of numerical weather models relate to errors that arise from model imperfections, assumptions, and approximations in microphysical parameterisations, although these factors have not been investigated here in detail.

2.2. Radar data-based methods

Radar reflectivity (Z) data can be used to derive spatially detailed rainfall rates (R) using the well known Z-R relationship, which can be converted into rainfall accumulations. This methodology does, however, have some problems, such as the radar sensing reflectivities that are not at the Earth's surface, a range dependence in the Z-R relationship, clutter removing algorithms removing actual rainfall data (Sansom *et al.*, 2001), and

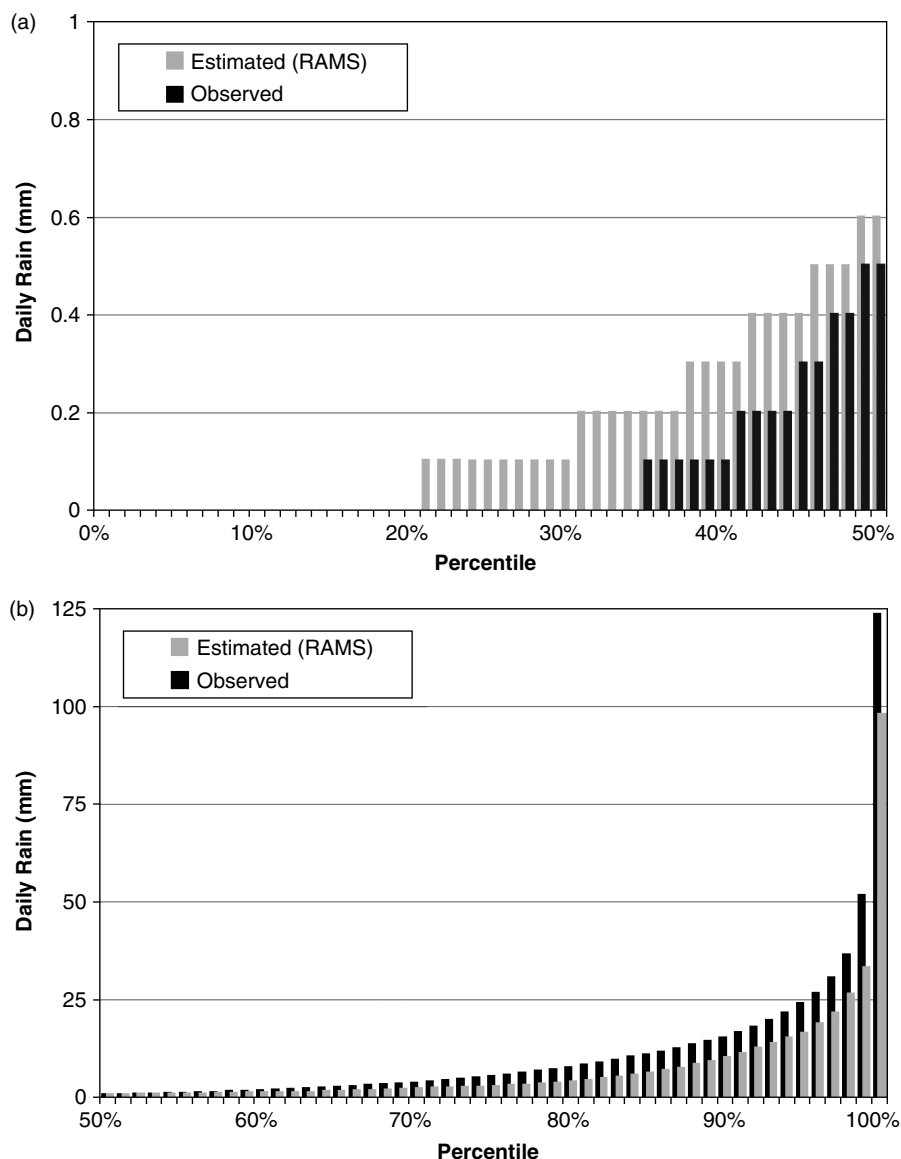


Figure 2. Probability distribution function (PDF) of daily rainfall from RAMS ($\Delta x = 20$ km, grey bars) and from rain gauge observations (black bars) for Raglan, New Zealand, for the period 1999–2003 for 0th–50th percentiles (a) and 50th–100th percentiles (b). See Figure 1 for the location of Raglan

introduction of considerable biases when merging data at one scale of resolution onto another scale, which would be the case when translating daily rainfall rates from radar onto a 0.05° latitude/longitude grid. For example, Steiner and Smith (2004) showed how merging a 5-min averaged raindrop spectra resulted in a 50% underestimation of mean and maximum 1-min rainfall rates. Additionally, it has been demonstrated that differences in the space and time resolution, overlap, and coverage of the area of observation can be significant enough that comparable estimates from gauges and radars could just be a matter of chance (Austin, 1987).

Even if the radar-derived estimates of rainfall were perfect, unfortunately, the coverage for which rainfall accumulations can be ‘reliably’ derived in New Zealand is limited to areas within 100 km of the three radar sites near Auckland, Wellington, and Christchurch (Figure 1). Additionally, radar data for these three areas

have only been available since the mid-1990s, and there are several temporal gaps in the records for each site. These facts combined rule out radar-derived estimates as a feasible option for this study.

2.3. Satellite data-based methods

Satellite data-based methods offer a key advantage over radar data-based methods for New Zealand as they are not limited in spatial coverage. Also, recently described methods (Bellerby, 2004) suggest improved algorithms (based on an Artificial Neural Network (ANN) approach using infrared (IR) data from geostationary satellites) that may overcome some of the problems with deriving rainfall accumulation estimates from instantaneous snapshots. However, while the method of Bellerby (2004) was a significant improvement over previous methods, the performance of this method (a correlation of 0.31 with 15 km, 15-min ground truth estimates) was insufficient to justify its operational use.

Geostationary satellite-based methods (using IR data) are restricted to inferring rates of precipitation from the presence of clouds. Microwave sensors on board polar orbiting satellites (such as the Tropical Rainfall Measuring Mission (TRMM) and NASA's Terra and Aqua platforms), on the other hand, can be used to estimate precipitation rates directly as microwave radiation is responsive to the presence of hydrometeors associated with the precipitation process. However, polar orbiting satellites are limited in that the temporal sampling of any location on the Earth is only two or three times a day, which results in an under-sampling of rainfall events at the daily timescale. Blended products that utilise the high temporal sampling of geostationary satellites and the more direct physical relationship between microwave radiation and precipitation have been developed (Adler *et al.*, 1994; Kidd, 2001), although reliable merged data sets only go back as far as 1987 (when passive microwave data from the Special Sensor Microwave Imager, SSM/I, became available) and long-run global data sets such as that from the Global Precipitation Climatology Project (GPCP) only have precipitation data at the monthly time scale and at 2.5° latitude/longitude grid resolution.

Advanced methodologies and improved deployment of satellite instruments offer the promise of further improved space-based rainfall estimates in the future. These should be considered for purposes of comparison (using verification methodology as described in Ebert, 2002) with other spatially complete rainfall estimates for New Zealand. However, for the purposes of this study, satellite data-derived rainfall estimates are not adequate.

2.4. Spatial interpolation methods

The underlying usefulness of spatial interpolation methods is directly related to the quality and quantity of the surface observations. In New Zealand, as has been mentioned in Section 1, there is an excellent and extensive network of currently open climate stations that record rainfall totals every day. Historically, the network has been more than double the current number of stations (the first meteorological observations for New Zealand were made in 1852 and a maximum of 1652 open rainfall stations was reached in October, 1970); however, this was in an era before government funding levels required a rationalisation of the network. Nevertheless, unlike the radar- and satellite-based methods described above, there are sufficient observational data for generating historic (1960–2004) daily rainfall estimates for the entire country. In addition, access to the data is very efficient. Most of New Zealand's rainfall data are stored in the National Climate Database (CLIDB), an Oracle relational database. This is maintained by the National Institute of Water and Atmospheric Research (NIWA) in Wellington, New Zealand. Data on daily rainfalls are quality checked and stored as 24-h totals from 9 to 9 A.M., New Zealand Local Time (NZLT).

There are several spatial interpolation models that have been used to interpolate climate data from surface observations covering large areas and encompassing complex terrain. The most common models are inverse distance weighting, Gaussian weighting, trend surface analysis (including linear and polynomial regression), kriging (including cokriging), and splines (Borga and Vizzaccaro, 1997; Daly *et al.*, 2002; Hutchinson and Bischof, 1983; Hutchinson and Gessler, 1994; Laslett *et al.*, 1987; Matheron, 1981; Philips *et al.*, 1992; Saveliev *et al.*, 1998; Seaman and Hutchinson, 1985; Thornton *et al.*, 1997). In this study we have chosen

a thin plate smoothing spline model, which has been shown to perform well in New Zealand conditions (Sansom and Thompson, 2003; Tait and Turner, 2005; Zheng and Basher, 1995).

A thin plate smoothing spline works by fitting a surface to the data with some error allowed at each data point, so the surface can be smoother than if the data were fitted exactly. Each station is omitted in turn from the estimation of the fitted surface and the mean error is found. This is repeated for a range of values of a smoothing parameter, then the value that minimises the mean error is taken to give the optimum smoothing. This process is called minimising the *generalised cross validation* (GCV). It can be automated once the order of the derivative, which controls the surface roughness, has been chosen.

The spline computation runs quickly, taking approximately 6 s to interpolate 1 day's rainfall data from 500 climate stations onto a 0.05° latitude/longitude grid covering all of New Zealand (11 491 grid points) on a UNIX machine with 2 GB of RAM. Thus, unlike using a NWP model, generating daily spline interpolations for several decades is not computationally inhibitive.

The density of the observation network is the primary deficiency for interpolation methods. In lowland areas with relatively uncomplicated terrain (i.e. where most of the population reside) there is an abundance of rainfall data. On the other hand, rainfall observations in areas of mountainous terrain are often sparse and records are often relatively short with many missing values. This is not uncommon for many areas around the world. Interpolation of rainfall data into these areas based on data from stations at lower elevations sometimes several kilometres away can cause significant discrepancies. It is also possible that some localised rainfall events may be missed entirely if no rain fell at the nearest observation stations, although this situation has been shown to be relatively rare in New Zealand (Turner and Tait, 2005).

The next section shows the results of an initial exploratory data analysis and Section 4 discusses and evaluates two different spline model parameterisations, bivariate (i.e. using two position variables only) and trivariate (i.e. using two position variables and elevation).

3. INITIAL EXPLORATORY DATA ANALYSIS

The pattern of rainfall on a daily basis for New Zealand is highly dependent on the passage of frontal systems and depressions over or near the country. This is clearly shown in Figure 3, which is a coloured contour plot of the rainfall for the 24-h period from 9 A.M. NZLT on the 27th of September 2003 to 9 A.M. NZLT on the following day. The plot was derived from 595 rainfall observations throughout the country. Also shown in Figure 3 is the synoptic chart valid at 6 A.M. NZLT on the 28th of September, 2003.

The depression and associated cold front situated to the west of the country (Figure 3) resulted in significant rainfalls for the western North Island and northwestern South Island. These areas were highly exposed to the prevailing northwesterly winds that were heavily laden with moisture. In addition to wind exposure, there is also evidence from Figure 3 that the rainfall totals for this 24-h period were enhanced by orography. The high rainfall area in the North Island extends inland and south, following the high relief areas shown in Figure 1. There was also significant rainfall down the west coast of the South Island, following the line of New Zealand's principal mountain chain, the Southern Alps.

To test the influence of orography on the daily rainfall totals for this day, a scatter plot was produced comparing elevation above mean sea level with the rainfall total recorded at the 595 observation sites. Figure 4 shows this scatter plot, which indicates that while there is a slight positive relationship, it is not statistically significant. The mean daily rainfall total for all of September 2003 at all the observation sites was also related to elevation; however, this relationship is also not statistically significant despite a slight improvement to the correlation (Figure 5).

4. THIN PLATE SMOOTHING SPLINE INTERPOLATIONS

On the basis of a comparison of the methods listed in Section 2, it was decided to use thin plate smoothing spline interpolations to produce multi-year (1960–2004) spatial estimates of daily rainfall for New Zealand on a 0.05° latitude/longitude grid. The software used was ANUSPLIN version 4.2 (Hutchinson, 2005). Bivariate

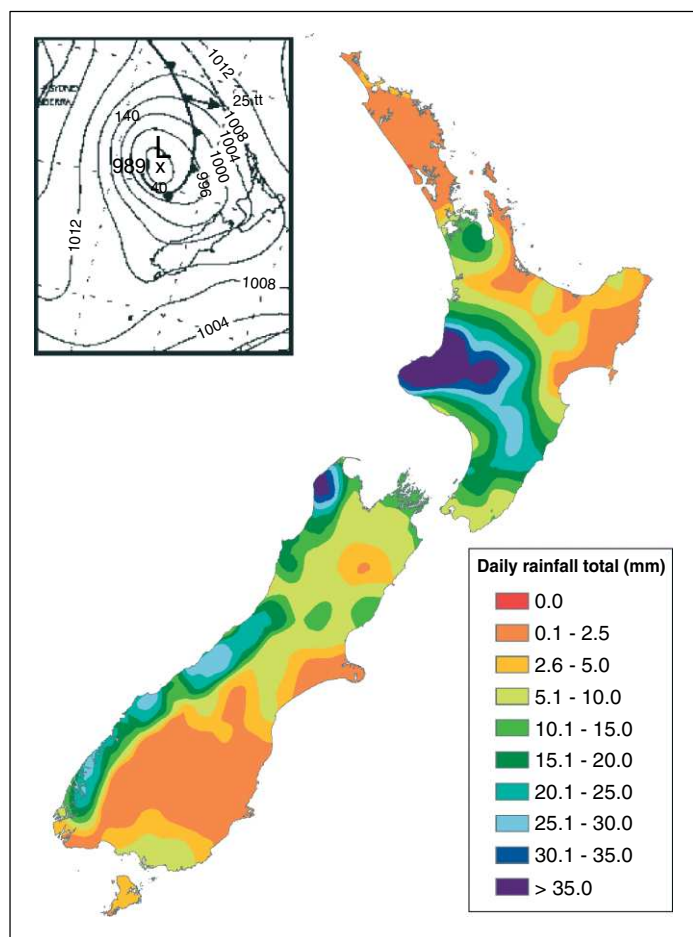


Figure 3. Coloured contour map of rainfall for the period 9 A.M. (New Zealand Local Time, NZLT) September 27th 2003 to 9 A.M. (NZLT) the following day. Also shown is the synoptic weather situation map valid at 6 A.M. NZLT September 28th 2003

(latitude and longitude) and trivariate (latitude, longitude, and elevation) spline interpolations were performed on a subset of 128 climate stations that had complete and reliable daily rainfall records for a two-year test period (September 2001–August 2003) and whose locations were well spread over the country.

The interpolations were performed both with and without an *a priori* square root transformation of the daily rainfall data, and the results were evaluated using a ‘leave one out’ methodology. This involved removing one station from the set, interpolating the daily rainfall from all the remaining stations to the omitted station’s location for each day of the two-year period, and then calculating the root mean square error (RMSE) from the estimated and actual values. The process was then repeated for each of the remaining 127 stations.

4.1. Bivariate spline interpolations

Hutchinson (1998a) describes the use of a bivariate (or two-dimensional) thin plate smoothing spline model to interpolate daily rainfall values from 100 sites. The two independent spline variables used were northing and easting location variables. The degree of smoothing was determined using a second order partial derivative by minimising the GCV. RMSE values were calculated by comparison of estimated and interpolated values (both with and without an *a priori* square root transformation) at 367 withheld data points.

Results from the Hutchinson (1998a) study showed that a square root transformation (and subsequent back transformation) yielded more accurate estimates of the withheld data values than directly interpolating the

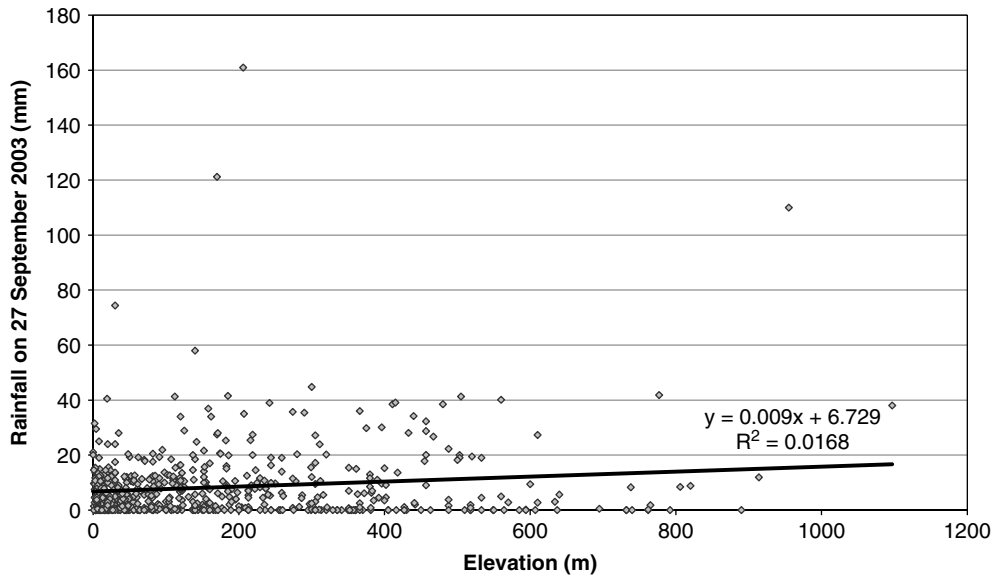


Figure 4. Scatter plot showing 24-h rainfall total for the period 9 A.M. (NZLT) September 27th 2003 to 9 A.M. (NZLT) the following day measured at 595 locations throughout New Zealand *versus* elevation above mean sea level. The solid line is the linear fit to the data. The regression equation and R^2 are also shown

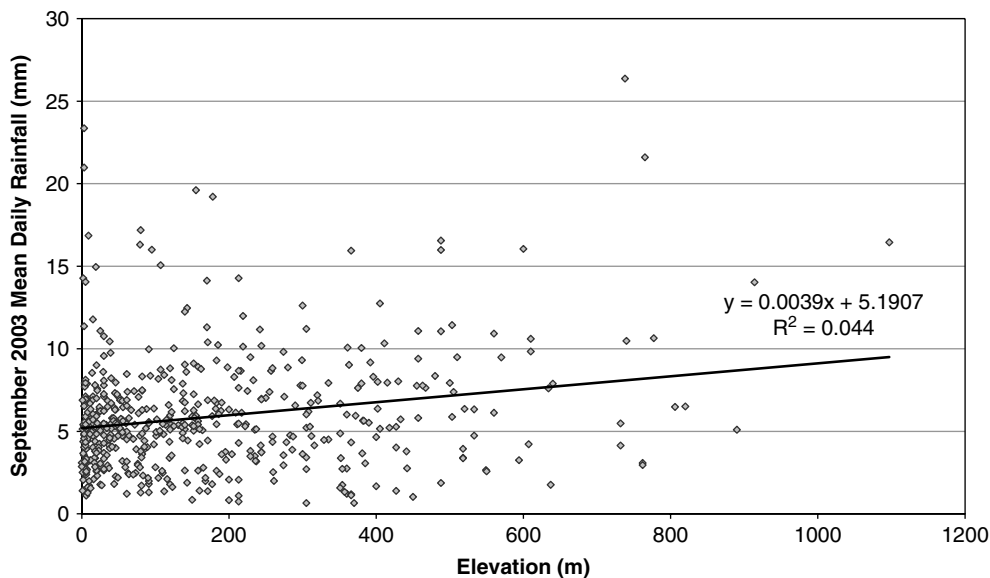


Figure 5. Scatter plot showing the mean daily rainfall total for September 2003 measured at 595 locations throughout New Zealand *versus* elevation above mean sea level. The solid line is the linear fit to the data. The regression equation and R^2 are also shown

untransformed values. The author concludes this is due to a reduction in the skewness of the data, which results in estimated errors that are positively correlated with the rainfall values. A third order partial derivative model was also experimented with but its performance was poorer than the second order analysis, which Hutchinson suggests is due to surface climate data rarely exhibiting the high degree of spatial continuity implied by third and higher order thin plate splines.

In the present study, a similar two-dimensional spline model was used (latitude and longitude as the independent variables, a second order derivative for the smoothing parameter, and minimisation of the

GCV) for interpolating the 2 years of daily rainfall data. As with the Hutchinson study, daily rainfalls were interpolated both with and without an *a priori* square root transformation (and subsequent back transformation). The RMSE, averaged over all 128 stations, for the untransformed rainfall data interpolations was 13.8 mm. In comparison, the average RMSE for the square root transformed rainfall data was 6.9 mm. A Student's *t*-test (Wonnacott and Wonnacott, 1982) comparing the untransformed and transformed RMSE values from each station showed that the two data sets are significantly different ($p < 0.001$). Thus, for our test data set, a square root transformation (and subsequent back transformation) significantly reduces the daily rainfall interpolation error.

4.2. Trivariate spline interpolations

Most meteorological variables, including rainfall, are affected by orography. From the initial exploratory data analysis performed for this study (Section 3), it was determined that there is likely to be some relationship between New Zealand's orography and rainfall, though it is unlikely to be simple and linear. Thus, since a spline interpolation is a non-linear fit of a surface generated by minimising the interpolation error, it makes sense to interpolate rainfall using a spline model with two position variables and an orographic variable such as elevation. The broad spatial pattern is determined by the two position variables, while the inclusion of elevation modifies the broad pattern to give more precise representations of the higher resolution variability. Hutchinson (1995) used a trivariate thin plate smoothing spline (latitude, longitude, and elevation) to interpolate mean annual rainfall (and other meteorological variables) across Australia. A trivariate spline, which allows the relationship between rainfall and elevation to vary spatially, was deemed more appropriate for a continent-wide interpolation, compared with a trivariate partial spline (including a single linear dependence on elevation), which is more suited to small-scale applications.

The orographic dependence of daily rainfall was investigated by Hutchinson (1998b). Several spline models were experimented with (bivariate, trivariate partial, and trivariate), which included two models with north–south and east–west aspect variables. Results showed that there were small, but significant, elevation and orographic aspect effects in the daily rainfall data. In the present study, a second order derivative trivariate (latitude, longitude, and elevation) thin plate spline (minimising the GCV) was used to interpolate daily square root transformed rainfall for the 2-year test period at the 128 sample stations. The RMSE, averaged over all the stations, was 4.9 mm. A Student's *t*-test showed that the difference between RMSE values derived from the bivariate and trivariate splines (both interpolation runs using square root transformed rainfall data) is significant ($p = 0.002$). Thus, we can conclude that a trivariate spline incorporating a square root transformation should be used for daily rainfall interpolations in New Zealand.

4.3. Validation using river flow data

An interpolated daily rainfall data set for all of New Zealand on a 0.05° latitude/longitude grid for the period 1960–2004 was produced using the trivariate spline model described in Section 4.2. River flow records from throughout the country were used to validate these rainfall data by comparing the estimated average annual rainfall (based on the daily estimates over the whole period) with long-term measured river flows plus estimated actual evapotranspiration (ET). Thompson *et al.* (1997) used a similar method of average annual rainfall validation using river flow data for the southern region of the North Island of New Zealand.

Past analyses that have used New Zealand's country-wide river flow data have used the full record at each site as flow records in New Zealand are generally short (McKerchar and Pearson, 1989; Pearson, 1995). In order to provide a basis for comparison with the rainfall data set, it was decided to normalise the flow data to the same period as the rainfall data set (1960–2004). Monthly flow data from 345 river flow records were examined for missing months. Missing values, either within a record or at either end of a record, were filled by correlation. This was done by testing the correlation of each record with all the others that have contemporaneous data. Each missing month was filled using a correlation against the same month at other sites, thus preserving the seasonal behaviour. Correlating flow records were chosen for the lowest standard error of estimate, penalised by a factor related to inter-site distance (linearly varying from 1 for inter-gauge distances up to 100 km, to 2 for inter-gauge distances of 500 km or more).

River flow values for each catchment were then converted from units of flow (m^3/s) to runoff (mm/yr) by dividing by the catchment area (mm^2) and multiplying by the number of seconds per year.

Validation of rainfall data using river flow data requires an estimation of the actual ET. In turn, an estimation of the long-term average annual actual ET requires an estimation of the long-term average annual potential evapotranspiration (PET). PET was estimated by spatially interpolating (using a trivariate spline with latitude, longitude, and elevation as independent variables) values of average annual Penman PET (Penman, 1948) at 215 climate stations located throughout New Zealand. At most climate stations, the Penman value was calculated using standard meteorological data, but for some stations without all the required meteorological observations the Penman value was estimated using either Priestley–Taylor PET or pan evaporation (with the appropriate bias corrections).

The actual ET estimation method of Zhang *et al.* (2004) was then applied to the rainfall and PET estimates. It defines an envelope of actual over potential, depending upon a catchment parameter w . The parameter w has been fitted by least squares to the output from the water balance model of Porteous *et al.* (1994) at 18 climate stations throughout New Zealand to give a value of 4.35. This compares with a range of global values between 1.7 and 5.0 (Zhang *et al.*, 2004).

Runoff, ET, and rainfall are not the only components of the water balance of a catchment. Water is also held in storage in the vegetation canopy, the soil, rivers, lakes, snowfields, glaciers, and deep groundwater. If runoff quantities are integrated over a number of years, then the average effect of many of these transfers becomes small in relation to the total amounts of water, and net water storage changes can be assumed negligible over an averaging time of 15 years or more. Losses to deep groundwater and storage in glaciers are significant in parts of New Zealand, but have significant effects on only a few of the flow records used in this study.

Thus the water balance reduces to

$$P - E - R - E_r = 0 \quad (1)$$

where P = precipitation

E = evapotranspiration

R = runoff

E_r = error term

To display the rainfall validation results, the error term is mapped, but is first normalised by the assumed known part of the water balance, and expressed as a fraction

$$E_r = (P - (R + E)) / (R + E) \quad (2)$$

Integrating the average annual rainfall and ET surfaces over 345 monitored catchments allowed calculation of the error term in Equation (2) above. Figure 6(a) shows the mapped error term (as a percentage), and bias and RMSE statistics are presented in Table I (model 1). From Figure 6(a) it can be seen that the error term is typically negative for most of the catchments analysed and is often greater than -25% . This is particularly so over the south and east of the North Island and most of the South Island. Some high elevation catchments in the Southern Alps show an underestimation of annual runoff of more than 50% . Some of this error will be attributed to the estimation of PET and some to the estimation of the average annual rainfall. As will be seen in the following section, much of this error can be reduced by replacing elevation with a mean annual rainfall surface as the third independent variable in the rainfall interpolation spline model.

5. INCLUSION OF A MEAN ANNUAL RAINFALL VARIABLE

From the hydrological model validation study of the long-term average annual rainfall described in Section 4.3, it was identified that the interpolated rainfall data (based on a trivariate spline using elevation) in the mountainous areas of New Zealand underestimated the actual rainfall by up to 75% . An approach by which

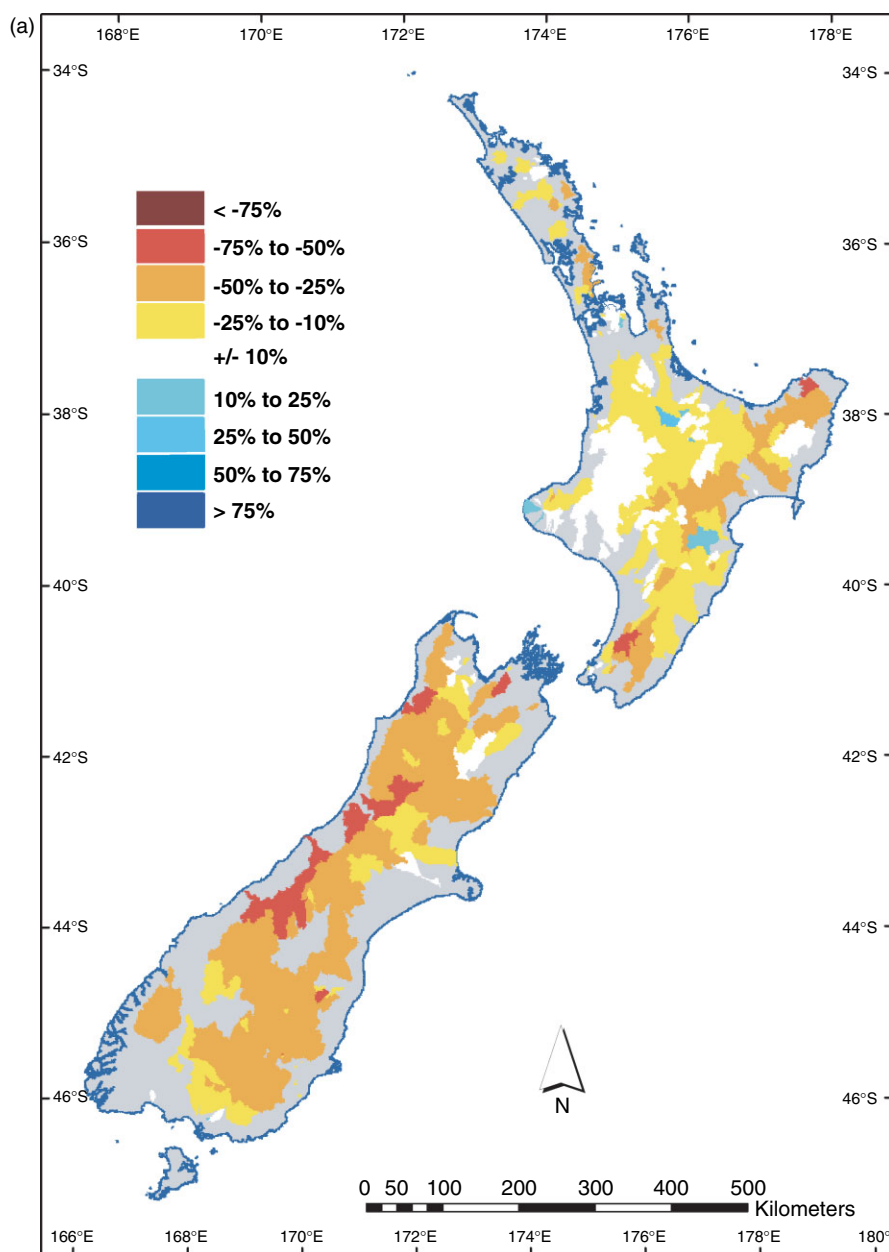


Figure 6. (a) Map of the New Zealand catchment areas analysed in this study showing the error term from Equation (2) (as a percentage) using average annual rainfall derived from a trivariate spline model with latitude, longitude, and elevation as independent variables. Grey areas on the map were not analysed. (b) Map of the New Zealand catchments analysed in this study showing the error term from Equation (2) (as a percentage) using average annual rainfall derived from a trivariate spline model with latitude, longitude, and the 1951–1980 mean annual rainfall as independent variables. Grey areas on the map were not analysed

the interpolated values in high elevation areas were scaled using limited high-resolution RAMS runs improved the results marginally (Tait and Turner, 2005). Testing this RAMS-adjusted annual rainfall surface against measured runoff (as in Section 4.3 above) showed that while the bias was almost removed the standard deviation was increased, because the North Island became almost entirely over-estimated while the South Island remained underestimated (Table I, model 2).

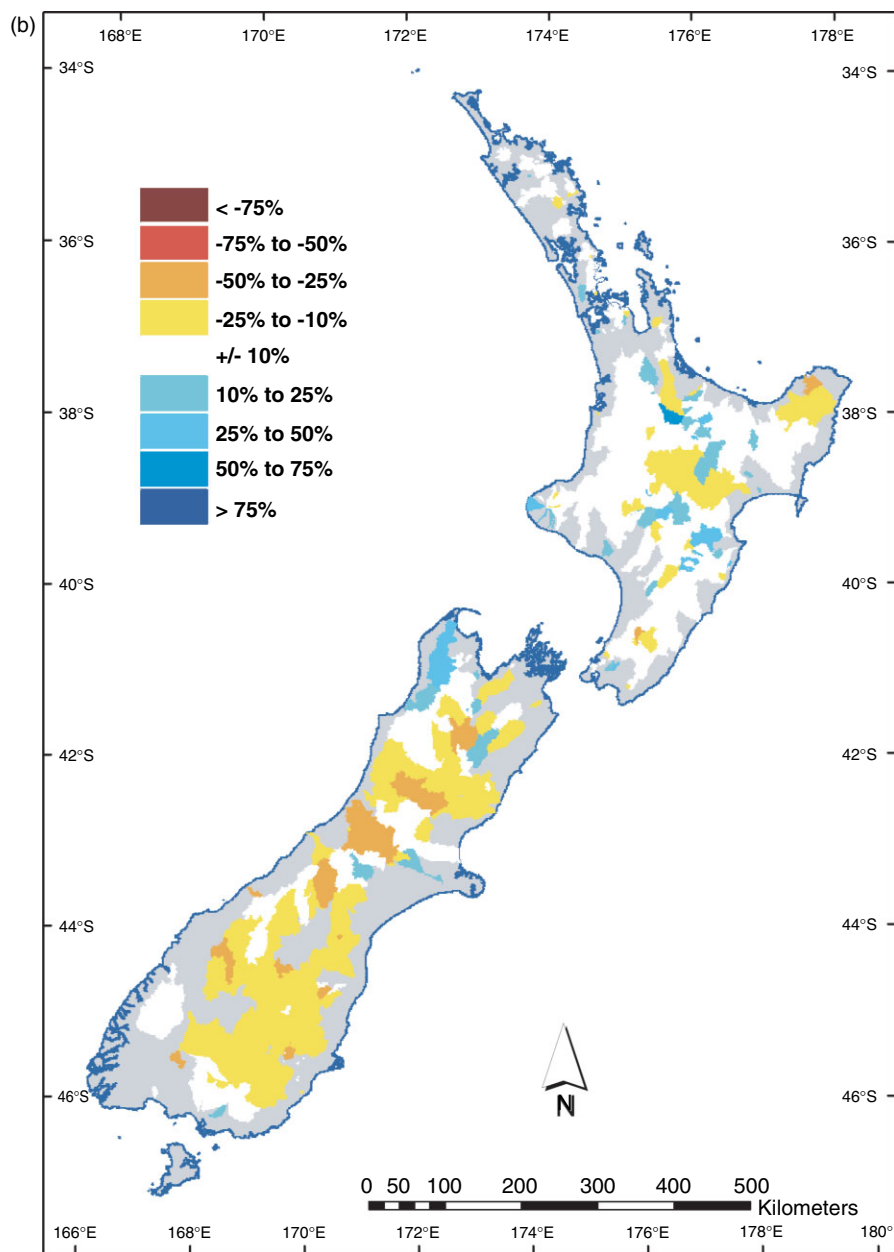


Figure 6. (Continued)

To improve the interpolations it was proposed that a digitised version of a hand-drawn map of the mean annual rainfall for the period 1951–1980 be used (the derivation of the 1951–1980 map is described in Section 5.1), instead of elevation, in the trivariate spline model. The hypothesis was based on the observation that the 1951–1980 map showed more realistic rainfall totals in the mountains (based on short-term observations, some high-resolution NWP model runs, and observed river flows) than those produced from the elevation-based spline model.

As a further test, the scatter plots shown here in Figures 4 and 5 were reproduced using the 1951–1980 mean annual rainfall values rather than elevation (Figures 7 and 8). The relationship between the daily rainfall totals for 27 September 2003 is improved as compared with that using elevation, though the correlation is not

Table I. Area-weighted bias and RMSE for estimation of rainfall from four different models, when tested against runoff and evapotranspiration estimates. Values are given for all of New Zealand and each island. See text for details of methods

Model	New Zealand		North Island		South Island	
	Bias (%)	RMSE (%)	Bias (%)	RMSE (%)	Bias (%)	RMSE (%)
1. Spline + elevation	−26	30	−17	21	−34	36
2. Spline + elevation + RAMS	1	33	16	38	−11	28
3. 1951–1980 mean annual rainfall	−6	14	−2	12	−8	16
4. Spline + 1951–1980 rainfall	−7	15	−1	12	−11	17

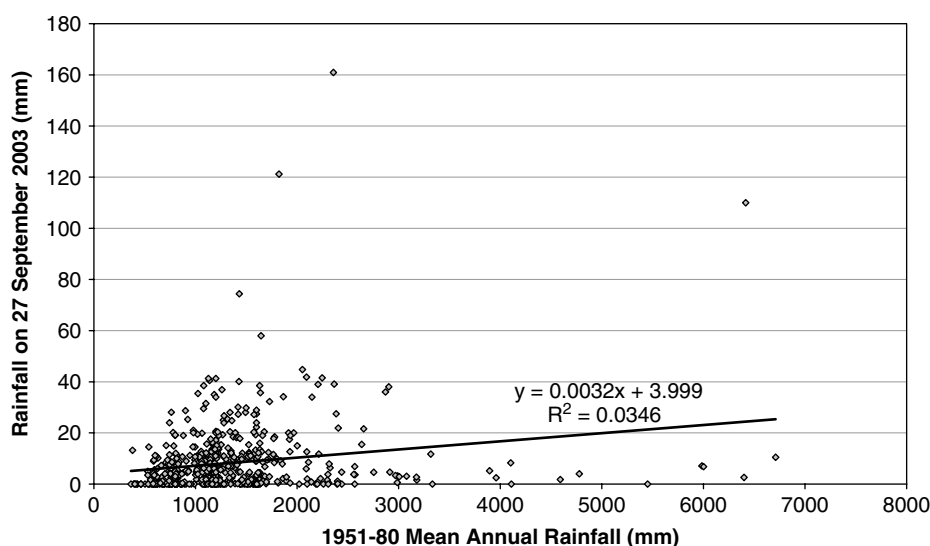


Figure 7. Scatter plot showing 24-h rainfall total for the period 9 A.M. (NZLT) September 27th 2003 to 9 A.M. (NZLT) the following day measured at 595 locations throughout New Zealand *versus* the 1951–1980 mean annual rainfall total. The solid line is the linear fit to the data. The regression equation and R^2 are also shown

statistically significant. However, the relationship between the mean daily rainfall for September 2003 and the 1951–1980 mean annual rainfall is statistically significant and much improved over that using elevation. This indicates that using the 1951–1980 mean annual rainfall surface in the daily rainfall interpolations rather than using the elevation may improve the accuracy of the daily rainfall estimates.

5.1. Derivation of the 1951–1980 mean annual rainfall map

The New Zealand Meteorological Service produced a climatic map series (1:2000 000 scale for all of New Zealand) in the 1980s, which included a map of the mean annual rainfall total based on data from the period 1951–1980 (New Zealand Meteorological Service, 1985). The hand-drawn contour map was based on observations at climate stations, with an expert interpolation by visual observation of rainfall for locations with few or no observations (J. Sansom, NIWA, personal communication). This included most of the mountainous areas of the country, the remote areas of the southwest of the South Island, and some central North and South Island locations. Recently, the hand-drawn contours were digitised and converted into vectors, which were then interpolated onto a 1 km raster grid.

The 1951–1980 mean annual rainfall surface was also subjected to testing against measured runoff less the ET, and showed a significant improvement over the two previous rainfall surfaces discussed above (Table I, model 3). It is thus a promising candidate as an independent variable for interpolating daily rainfall.

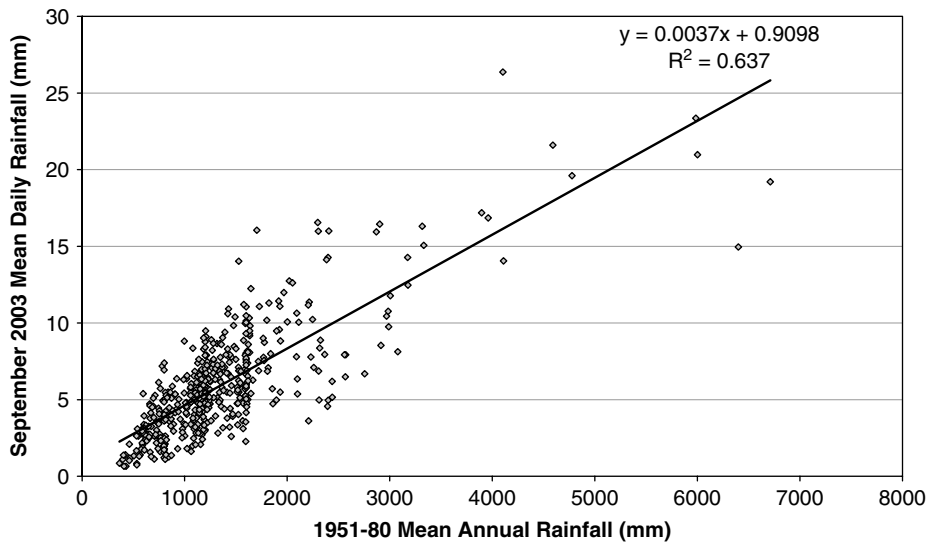


Figure 8. Scatter plot showing the mean daily rainfall total for September 2003 measured at 595 locations throughout New Zealand versus the 1951–1980 mean annual rainfall total. The solid line is the linear fit to the data. The regression equation and R^2 are also shown

5.2. Three-dimensional spline interpolations

The daily rainfall interpolations described in Section 4.2 were repeated using the 1951–1980 mean annual rainfall surface, rather than elevation, as the third variable in the trivariate spline model. It is argued that the pattern present in the mean rainfall map is a better representation of the spatial distribution of daily rainfall in New Zealand, compared with elevation. A criticism of this approach is that not all rainfall events will conform to the same general spatial pattern (i.e. a winter-time southerly frontal storm will produce a different spatial pattern over the country than a summer-time convective system). This limitation of the method can be somewhat overcome by producing ‘expert-guided’ mean rainfall surfaces associated with a set of typical synoptic situations for New Zealand, such as those of Kidson (2000). However, at present these synoptic-dependent surfaces have not been produced.

Using data from the 128 sample stations, the average RMSE for the trivariate spline run using the 1951–1980 rainfall surface was 4.8 mm. A Student’s t -test showed that the difference between RMSE values derived from the bivariate and trivariate (using the 1951–1980 rainfall surface) splines (both interpolation runs using square root transformed rainfall data) is statistically significant ($p = 0.0005$). The difference between RMSE values derived from the two trivariate spline runs (using elevation and the 1951–1980 rainfall surface) is not significant ($p = 0.8849$).

Despite the latter result, it was considered that inclusion of the 1951–1980 rainfall surface in the trivariate spline model could produce improved daily rainfall estimates compared with using elevation, particularly in mountainous regions. As has been previously noted, the 1951–1980 mean annual rainfall map showed more realistic rainfall totals and patterns in the mountains than those produced from an elevation-based spline model. Comparison of the two mean annual rainfall surfaces at low elevations showed no major differences, which is the likely cause of the non-significant result above, as most of the 128 sample stations are at low elevations.

In the test data set, only 12 of the 128 stations are located above 500 m in elevation, and only 3 are located above 800 m (at 820, 825, and 881 m above sea level). The average RMSE values at the 12 stations above 500 m for the interpolations using the 1951–1980 rainfall surface is 4.4 mm and at the 3 stations above 800 m is 4.3 mm. By comparison, the same errors for the elevation-based interpolations are 4.5 and 4.8 mm. While the sample sizes are too small to statistically compare these averages, it can be seen that the difference between RMSE is greater (in favour of the 1951–1980 rainfall surface method) at the highest elevation sites.

To statistically test this result, a short-term high elevation rainfall data set, collected during an experiment in the Southern Alps in 1996, was analysed. This analysis is presented in the following section.

5.3. Comparison with a short-term high elevation data set

Three-hourly rainfall totals were observed at 21 high elevation sites (ranging between 655 and 1609 m above sea level) in an area of the Southern Alps between 41.1 and 44.1 °S during the Southern Alps Experiment (SALPEX) in October and November, 1996 (Wratt *et al.*, 1996). For this study, daily rainfall was interpolated to these high elevation sites for each day of the SALPEX period using the two trivariate spline methods described previously in Sections 4.2 and 5.2. Data from over 500 climate stations throughout the country were used for the interpolations. The SALPEX rainfall data were not included in the spline interpolations.

The RMSE, averaged over all the 21 SALPEX sites, for the elevation-based method was 24.6 mm (the error is much larger than that from the previous analysis owing to fewer observations, only 36 days, and a smaller sample size of stations). By comparison, the average RMSE for the 1951–1980 rainfall spline method was 18.6 mm. A Student's *t*-test showed that this difference is not statistically significant ($p = 0.315$). It is suggested, however, that the lack of significance is a result of the small sample size only, as visual comparisons of the spline-estimated rainfalls compared with the observed values consistently show the spline fit using the 1951–1980 mean annual rainfall surface to outperform the fit based on elevation data. Figures 9 and 10 are shown as typical examples.

Figure 9 shows the estimated and observed daily rainfalls at one of the SALPEX sites, Lake Ramsay (43.3°S, 170.9°E, 945 amsl; see Figure 1). It can be seen that while the timing of the rainfall events was well represented, the elevation-based spline method has substantially underestimated the observed rainfall. Over the 36 days of the SALPEX period, the total rainfall estimated from the elevation-based spline fit is only 59% of the observed total. In contrast, the total rainfall from 1951 to 1980 annual rainfall-based spline method is 114% of the observed total. Figure 10 shows the estimated and observed daily rainfalls at Tuke (43.1°S, 170.9°E, 975 amsl; see Figure 1). At this site, both the spline models significantly underestimate the

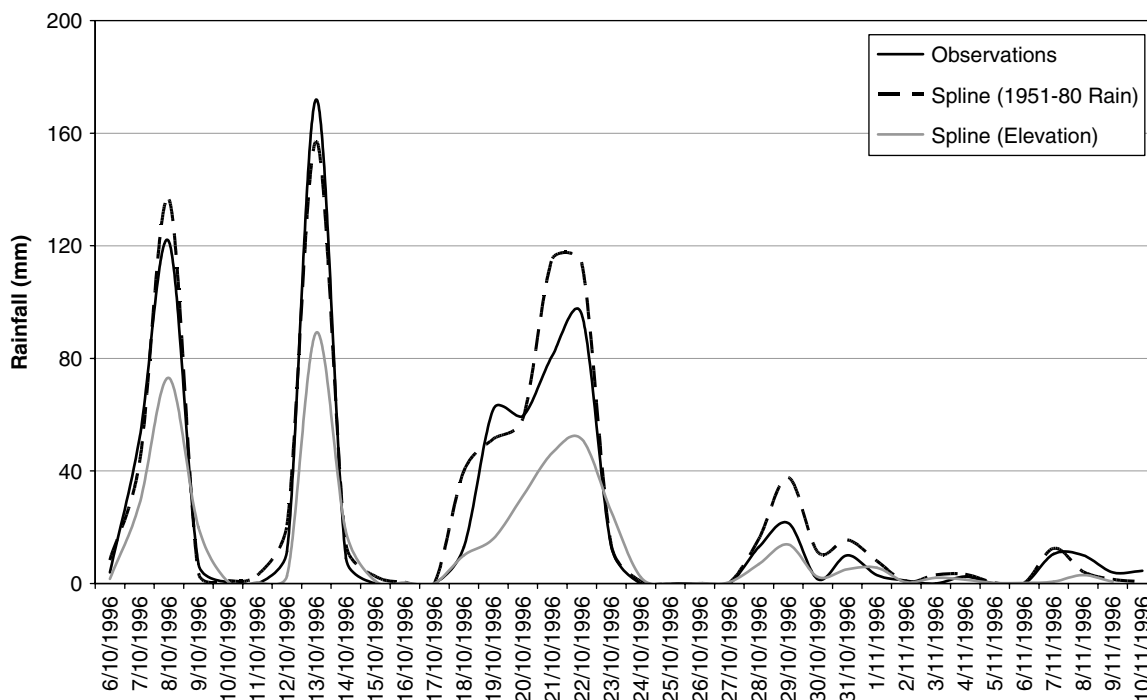


Figure 9. Comparison of observed and estimated daily rainfall at Lake Ramsay for the period 6th October–10th November, 1996. See Figure 1 for the location of Lake Ramsay

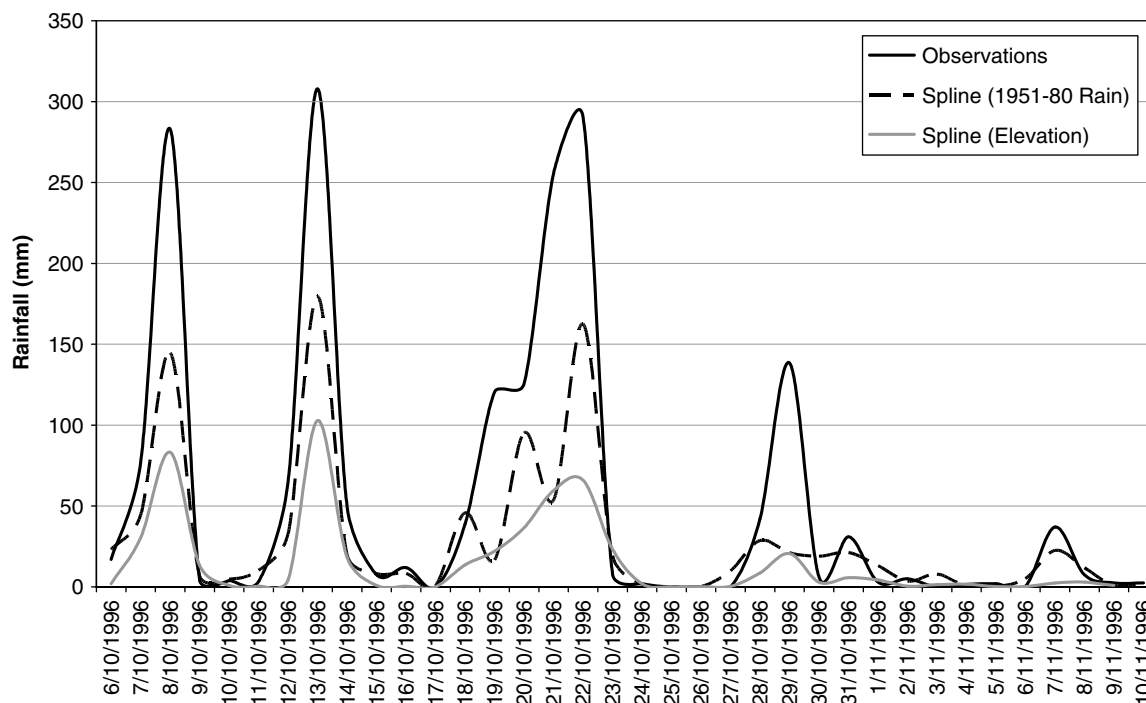


Figure 10. Comparison of observed and estimated daily rainfall at Tuke for the period 6th October–10th November, 1996. See Figure 1 for the location of Tuke

Table II. RMSE for the two trivariate spline model fits at each of the 21 high elevation SALPEx sites

Station name	Altitude (amsl)	Longitude (°E)	Latitude (°S)	RMSE (1951–1980 rainfall)	RMSE (elevation)
Dry acheron	655	171.66	43.40	12.38	13.21
NZDSA mathias	670	171.13	43.19	11.41	13.49
Griffiths	700	171.31	43.05	21.99	31.43
Elcho flats	739	169.84	43.92	7.62	7.59
Bull creek	745	171.97	42.90	5.33	4.15
Ball hut road bridge	759	170.13	43.74	13.50	5.72
Arthur's pass	760	171.56	42.94	7.05	12.45
The hermitage	765	170.09	43.73	10.78	15.22
Mistake flat	790	170.69	43.45	10.42	23.91
Nigger hill	790	172.05	42.99	16.30	17.27
Carrington	823	171.55	43.05	32.74	35.33
Trilobite	825	172.61	41.13	5.57	14.88
Lake Ramsay	945	170.92	43.28	9.89	21.49
Tuke	975	170.89	43.09	58.30	76.98
Waterfall	975	170.95	43.08	61.95	79.66
Eade hut	979	170.48	43.52	12.47	23.66
Cassinia moraine	980	169.67	44.12	8.93	10.86
Skifield	1364	170.66	43.83	5.03	9.25
Cabot	1430	171.27	43.05	50.30	62.18
Boanerges ridge	1447	169.79	43.99	2.66	2.74
Panorama ridge	1609	170.48	43.49	25.26	34.36

observations, but the 1951–1980 rainfall-based estimates are still an improvement over the elevation-based estimates (the percent of total observed rainfall is 27% for the elevation-based method and 54% for the 1951–1980 rainfall-based method).

Table II shows the RMSE values for each of the 21 SALPEX sites. It can be seen that at almost all locations the 1951–1980 rainfall-based spline estimate has a lower RMSE compared with the elevation-based spline estimate, particularly at the higher elevation sites. These results are consistent with the few high elevation stations in the 128-station test data set. Thus, on the basis of the above results, we believe the optimal method for interpolating daily rainfall in New Zealand is a second order trivariate (latitude, longitude, and the 1951–1980 mean annual rainfall surface) thin plate smoothing spline (minimising the GCV).

5.4. Validation using river flow data

The revised mean annual rainfall surface for 1960–2004 (using the 1951–1980 rainfall surface as an independent variable in the spline model) was subject to the same tests against measured runoff and ET. The error term from Equation (2) was once again mapped (Figure 6(b)). The results are very promising, showing a marked reduction (compared with Figure 6(a)) in the degree of estimation error throughout the country. Bias was also reduced and the standard deviation of the error was similar to that in the other models with the exception of the RAMS-adjusted surface. The model statistics are also shown in Table I (model 4).

6. DISCUSSION AND CONCLUSIONS

This study presents a method for estimating daily rainfall on a 0.05° latitude/longitude grid covering all of New Zealand for the period 1960–2004, for the purpose of providing rainfall data as an input for and testing of hydrological models. It is argued that the method of spatial interpolation of surface observations of daily rainfall, compared with NWP, radar data-based methods, and satellite data-based methods, is best suited to achieve this result. A thin plate smoothing spline model (ANUSPLIN) was chosen as this model has been shown to perform well in previous rainfall interpolation studies in New Zealand. Transformation of the daily rainfall data using a square root transformation (and subsequent back transformation) is shown to significantly reduce the interpolation error, as has the inclusion of elevation information in the spline model (i.e. trivariate vs bivariate schemes).

Further, through the analysis of a 2-year sample data set of daily rainfall observations from 128 climate stations situated throughout the country and a 36-day high elevation data set, we believe that the optimal spline model for estimating daily rainfall throughout New Zealand is a second order trivariate (latitude, longitude, and the 1951–1980 mean annual rainfall surface) spline (minimising the GCV). Some further improvements could be made by producing ‘expert-guided’ mean rainfall surfaces associated with a set of typical synoptic situations for New Zealand.

Interpolated daily rainfalls using the above ‘optimal’ spline model for the period 1960–2004 were validated by comparing the annual average rainfall for this period with average annual river flows and PET. This analysis showed that while there is still some error, particularly in the South Island catchments, the error has been substantially reduced compared with the previously tested models (a spline model using latitude, longitude, and elevation variables, and the same spline model with a RAMS-based scaling adjustment at high elevations).

The distribution and magnitude of the mapped error term from the ‘optimal’ model (Figure 6(b)) is similar to those when using the 1951–1980 mean annual rainfall in Equation (2). This result is also shown in Table I (compare models 3 and 4), and suggests that there are still some systematic errors in the 1951–1980 rainfall surface. One known discrepancy is the biased location of the 1951–1980 normal peak rainfall on transects across the Southern Alps southeast of Greymouth (see Chinn, 1979 and Henderson and Thompson, 1999 for transect profiles. See Figure 1 for the location of Greymouth). The 1951–1980 rainfall surface shows too much reliance on altitude, and fails to capture the extreme steepness of the rainfall gradients in this region. Results from the SALPEX programme (Wratt *et al.*, 1996) could allow a systematic correction to be applied; however, manual attempts to date have met with mixed success (A.I. McKerchar, NIWA, personal communication).

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